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Case Report

Physicochemical and mechanical properties of polyvinylidene fluoride nanofiber membranes

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A R T I C L E I N F O

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A B S T R A C T

Polyvinylidene fluoride (PVDF) is hydrophobic, piezoelectric, and has good chemical stability and high mechanical properties; therefore, it is often used to reinforce other composite materials for various applications. However, the characteristics of pure PVDF nanofibers have rarely been comprehensively explored. This study prepared pure PVDF nanofiber membranes using electrospinning and investigated their physicochemical and mechanical properties. Scanning electron microscopy revealed that the PVDF nanofiber surfaces were white, homogeneous, flat, bead-free, and smooth, and that the nanofiber diameters were 230–855 nm. Fourier-transform infrared analyses showed that the β phase was dominant in all the nanofibers. The spectra showed that the α, β, and γ phases were present in PVDF1 and PVDF2, whereas PVDF3 and PVDF4 were dominated by the β and γ phases. The PVDF nanofiber samples had a semi-crystalline structure with crystallinity values of 58.45–52.67%. The tensile strength of the PVDF nanofibers increased from 3.61 to 6.00 MPa with an increasing PVDF concentration. The application of a high voltage during the electrospinning process did not damage the chemical structure of the PVDF nanofibers. The PVDF nanofiber membranes produced in this study are potential candidates for application in wound healing, water filtration, and masks.

1. Introduction

Nanofibers can be easily and conveniently prepared using electrospinning [1,2]. Electrospinning is regarded as the most powerful and versatile method for manufacturing economically viable and adaptable nanoscale fibers [2,3]. This technique employs a high voltage to elongate a polymer solution. A high voltage induces the motion of charges inside the polymer liquid and facilitates the stretching of the charged droplets. When the electrostatic repulsion of the charged polymer liquid surpasses the surface tension, jet initiation results in a distinct cone called the Taylor cone [4,5]. The force leading to the formation of the Taylor cone can be indirectly controlled by the flow rate and applied stress, thus allowing the realization of a stable jet. Sufficient cohesive forces in the polymer fluid are essential to successfully stabilize the Taylor cone [6]. This ejection allows the polymer chains to stretch, eventually culminating in the formation of uniform fiber strands or beaded fibers.

The morphology, size, and surface of the fibers can be adjusted by

Abbreviations: PVDF, polyvinylidene fluoride; SEM, scanning electron microscopy; FTIR, Fourier-transform infrared; XRD, X-ray diffraction; PAN, polyacrylonitrile; DMF N, N-dimethylformamide; T, Trans; G, gauche.

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adjusting the electrospinning parameters, which consist of the solution parameters (molecular weight, polymer concentration, conductivity, viscosity, and surface tension) and process parameters (electrical voltage, needle-collector distance, and polymer solution flow rate) [7]. Due to the advantages and convenience of this technique, electrospun fibers are widely used in various applications, such as drug delivery systems [8], air filtration [9], tissue engineering [10], wound healing [11], and masks [12]. Natural and synthetic polymers are often used in the synthesis of nanofiber composites using electrospinning techniques [13,14]. Some synthetic polymers, such as polyacrylonitrile (PAN) [15], polypyrrole [16], polyamide-6 [17], polyethersulfone [18], polyvinylidene fluoride (PVDF) [19], and polyurethane [20], have been successfully used to develop nanofiber membranes [15] In particular, PVDF fibers have easily controllable hydrophobic structures, including high surface to weight ratio and small diameters (tens to hundreds of nanometers), and a piezo/piro/ferroelectric nature, which can be easily controlled by adjusting the production parameters. Therefore, electrospinning is a simple technique that can be used to produce PVDF nanofibers that can strengthen the mechanical properties and chemical resistance of PVDF nanofiber membranes.

Several studies have reported the successful fabrication of PVDF nanofiber composites with other materials using electrospinning. Shetty et al. developed a flexible nanogenerator using a PVDF/talc nanosheet composite. The addition of talc to the PVDF nanofibers increased the diameter, piezoelectricity, and tensile strength of the nanofibers from 278 to 396 nm, 1.6–9.1 V, and 10–45 MPa, respectively [21]. Chowdhury et al. reported the mechanical properties of Fe_3O_4-PVDF nanofibers [22]. Increasing the Fe_3O_4 concentration in the PVDF nanofibers decreased the nanofiber diameter to 77 nm and decreased the β phase quantity of the composite nanofiber mats, which decreased their piezoelectric response from 38 to 5% and decreased the crystallinity by 13.82% [22]. Teng et al. reported micro zein/PVDF nanofiber composites with improved mechanical properties. The combined micro zein and PVDF nanofibers exhibited a helical ribbon morphology with a nanofiber diameter of 2200 nm, a breaking stress of 0.73 MPa [23]and a better oil absorption performance of than those of rod fibers [23]. The aforementioned studies demonstrated the effects of other materials on PVDF. However, there have been few comprehensive discussions on the characteristics of pure PVDF nanofibers as fixed controls. Pure PVDF nanofibers were synthesized in this study using electrospinning, and the effect of increasing concentrations of pure PVDF on the physicochemical properties of the composites, including the morphology and chemical interactions, was analyzed. This included determining the phases present in the PVDF nanofiber composites to qualitatively assess the typical piezoelectric characteristics and changes in the crystallinity of PVDF. Furthermore, the mechanical properties of the fibers were evaluated using tensile testing.

2. Experimental methods

2.1. Materials

PVDF (MW: 58,000 kg/mol) and N, N-dimethylformamide (DMF) were obtained from Sigma-Aldrich (Singapore) and used in this study.

2.2. Preparation and synthesis of PVDF nanofiber membranes

PVDF nanofibers were prepared by dissolving various concentrations of PVDF, namely, 15% w/w (PVDF1), 20% w/w (PVDF2), 25% w/w (PVDF3), and 30% w/w (PVDF4), in DMF. Homogeneous PVDF precursors were produced by stirring the two materials for 1.5 h at 80 °C and 250 rpm using a digital stirrer. The PVDF1, PVDF2, PVDF3, and PVDF4 solutions were processed into nanofibers using an electrospinning machine (Nanolab ES/DS106, Malaysia), as shown in the schematic diagram in Fig. 1. Electrospinning was conducted at a flow rate of 3.33 μL/min, electric voltage of 13 kV, needle tip distance on the collector drum of 10 cm, and a collector drum rotation speed of 250 rpm. Moreover, electrospinning was conducted at a room temperature and humidity of 25 °C and 40–50%, respectively. Electrospinning was conducted for 4–5 h to produce nanofiber membranes.

2.3. Characterization of the PVDF nanofiber membranes

The morphologies of the PVDF1, PVDF2, PVDF3, and PVDF4 nanofibers were examined using scanning electron microscopy (SEM; Vega 3 Tescan, Japan). The diameters of the PVDF1, PVDF2, PVDF3, and PVDF4 nanofibers were measured 100 times using ImageJ version 1.52a (National Institutes of Health, Bethesda, MD, USA), and the distribution was analyzed using Origin 2018 (OriginLab Corporation, USA). Multiple diameter comparisons were performed using Tukey's honest significant difference post-hoc test. Insignificant, significant, and highly significant differences were determined using the p-values of >0.05, <0.05, and <0.01, respectively. The functional groups and crystallization patterns of the PVDF1, PVDF2, PVDF3, and PVDF4 nanofibers were determined using Fourier-transform infrared (FTIR) spectroscopy (Thermo Nicolet iS10, Japan) and X-ray diffraction (XRD) (Rigaku MiniFlex 600, Japan), respectively. The tensile strengths of the PVDF1, PVDF2, PVDF3, and PVDF4 nanofiber membranes were determined using a FAVIGRAPH tensile tester (Textechno H. Stein GmbH & Co., Germany) and the grasp and pull test method. All samples were cut into 3 mm × 20 mm rectangles of equal thickness, clamped on both sides of the apparatus, and stretched at an elongation rate of 20 mm/min. The tensile strength test results were analyzed using Origin software version 2018 (OriginLab Corporation, USA).

3. Results and discussion

3.1. Electrospinning of PVDF nanofiber membranes

The PVDF1, PVDF2, PVDF3, and PVDF4 nanofibers membranes were successfully produced using the electrospinning process. Images of the resulting nanofiber membranes are shown in Fig. 2. Macroscopic analyses revealed that the PVDF1, PVDF2, PVDF3, and PVDF4 nanofiber membranes exhibited similar surface characteristics, namely, smooth, non-brittle, white, and homogeneous surfaces that resembled the surface of tissue paper. Gee et al. reported similar findings, in which the

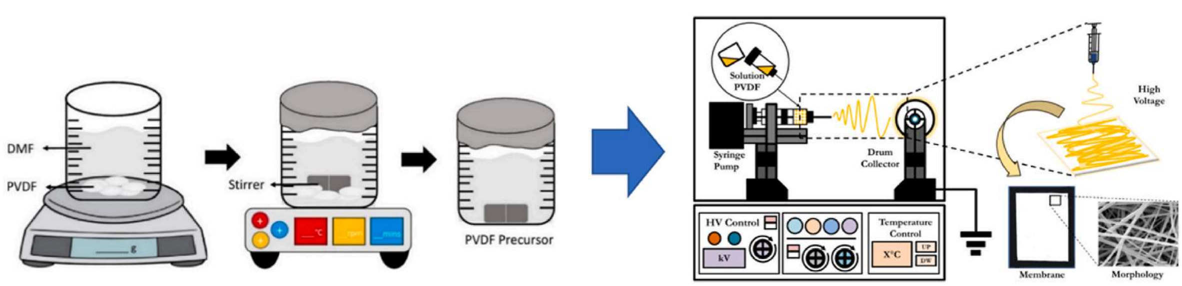

Fig. 1. Schematic illustration of the electrospinning process used in this study.

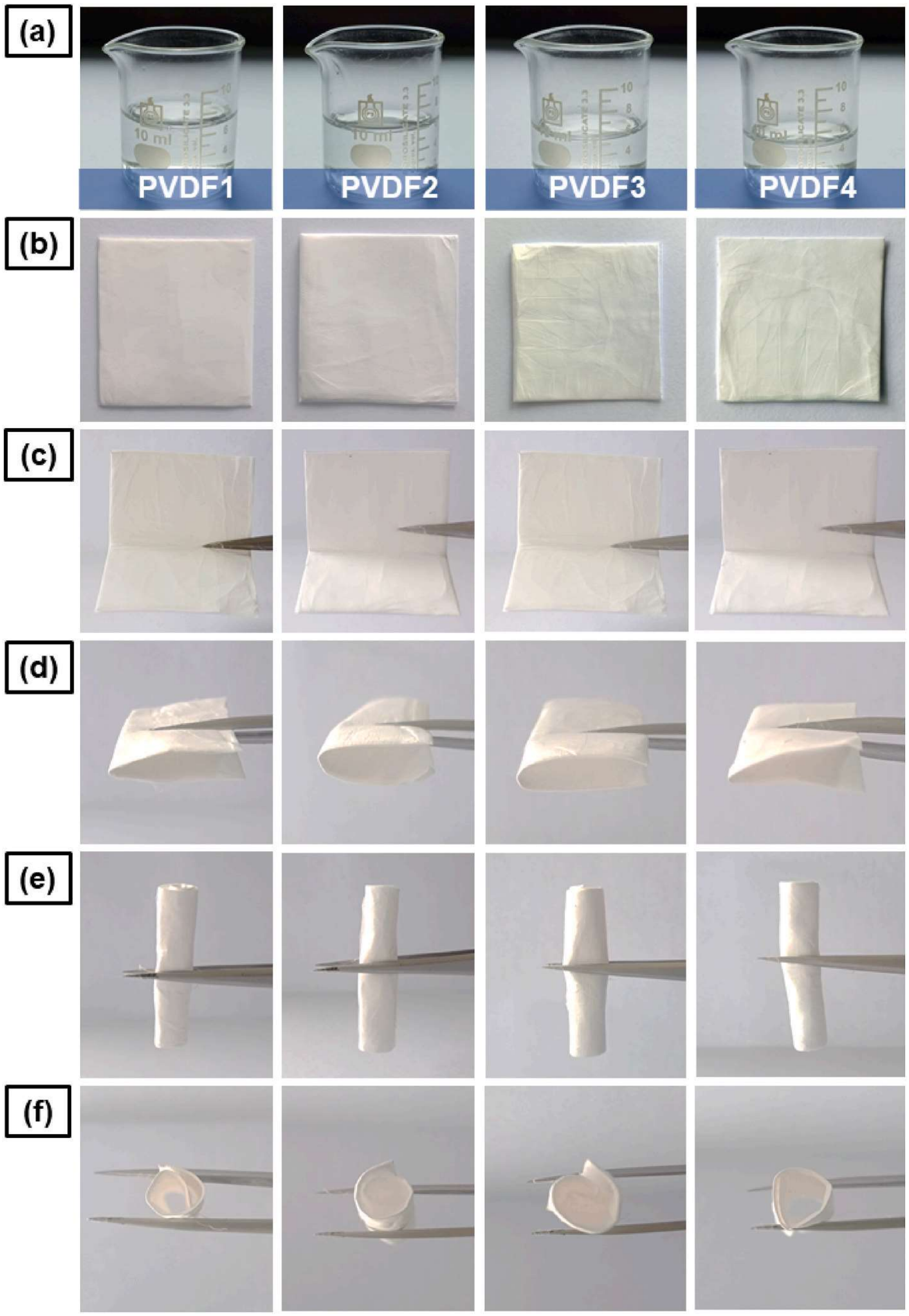

Fig. 2. Photographic images showing the (a) PVDF1, PVDF2, PVDF3, and PVDF4 solutions, (b) PVDF1, PVDF2, PVDF3, and PVDF4 nanofiber membranes, and (c, d, e, f) PVDF1, PVDF2, PVDF3, and PVDF4 nanofiber membranes bent at (c) 90, (d) 180, and (e, f) 360°.

resulting PVDF nanofibers exhibited homogeneity, a white color, and a smooth surface [24]. Connecting the positive electrode on the electrospinning needle to the negative electrode of a DC power source polarized the polymer solution owing to the potential difference between the electrodes, resulting in the successful formation of the PVDF1, PVDF2, PVDF3, and PVDF4 nanofibers [13,15]. Moreover, the success of the electrospinning process depends on several factors, including the Coulomb force, wherein a high-voltage electric current facilitates the smooth movement of the polymer through the tip of the needle towards the collector; the charge force, which provides PVDF/DMF solution droplets for spinning; and the surface tension force that occurs between air and the polymer to prevent dripping of the PVDF precursor from the needle tip [25,26]. The balance between these factors attracts the PVDF/DMF solution to create a Taylor cone [25,27]. A macroscopic analysis of the membranes revealed that all nanofiber membranes could be bent at angles of 90, 180, and 360°, as shown in Fig. 2(c–f), indicating that the PVDF1, PVDF2, PVDF3, and PVDF4 nanofiber membranes exhibit flexible properties. However, the membrane with smaller nanofiber diameters (PVDF1), as compared with that with larger nanofiber diameters (PVDF4), exhibited better flexibility properties. The results of this study are consistent with those of our previous studies, which demonstrated that PVP/CA and PAN/GO nanofibers are flexible at an angle of 90° [15,28].

3.2. Morphology of the PVDF nanofiber membranes

Fig. 3 shows the morphologies of the PVDF1, PVDF2, PVDF3, and PVDF4 nanofibers. The electrospun nanofibers were beaded, simple, and

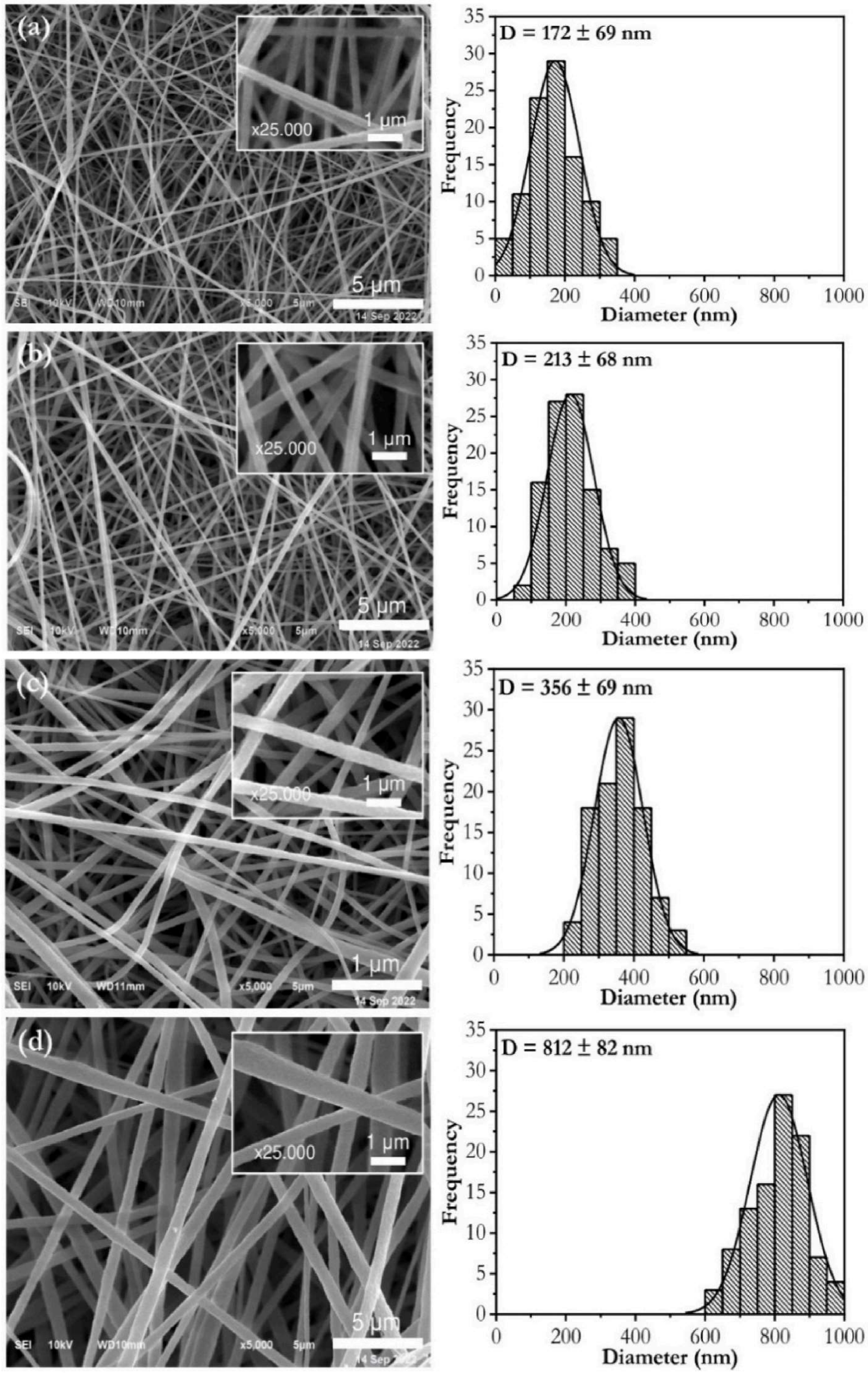

Fig. 3. SEM images and corresponding size distribution of the (a) PVDF1, (b) PVDF2, (c) PVDF3, and (d) PVDF4 nanofibers.

fine [29]. Similar observations were reported by Gee et al. [24] and Saha et al. [30]. Who found that PVDF polymers with concentrations of 20–30% w/w produced uniform fibers without beading [24,30]. This can be attributed to the delicate balance between the solution parameters, electrospinning process, and environmental conditions [31,32]. The SEM analysis results show that the diameter of the surface nanofibers increased with an increasing PVDF concentration, as well as the nano-PVDF, with the average diameters and standard deviation values of the diameter distribution of PVDF1, PVDF2, PVDF3, and PVDF4 being 172, 213, 356, and 812 nm, and 69, 68, 69, and 82 nm, respectively. The magnitudes of several factors, including the polymer concentration, voltage, and solution viscosity, influence the resulting diameter differences, with an increase in these factors correspondingly increasing the diameters of the nanofibers [33].

The results shown in Fig. 3 were compared with the diameter results from previous studies. As summarized in Table 1 the increase in diameter was directly proportional to an increase in the PVDF concentration. A comparison of the dependence of the diameter and concentration of the PVDF solution was performed using the slope regression value of the nanofiber diameter graph of the results of this study versus the diameter of the fibers reported by Motamedi et al. [34] and Shao et al. [34] as shown in Fig. 4(a) [34,35]. The slope regression values for all three

Table 1
The Diameters of PVDF nanofiber membranes.

Concentration (%)	Diameter (nm)	References
15	230	[35]
20	284	[35]
25	473	[34]
30	855	[34]

studies were the same at 206.03. This alignment indicates that the results of our study are consistent with the outcomes of earlier studies. However, the R^2 (coefficient of determination) value of our study was slightly better (0.8865) than those of the studies by Motamedi et al. [39] and Shao et al. [35] which exhibited an R^2 value of 0.8245. The degree of suitability of the results for the obtained data increased as the R^2 value approached unity [36].

The authors recently reported that increasing the concentration of PVDF, PVP, and PAN polymers increases the fiber diameter from 100 to 300 nm. However, the statistical analysis of the diameter distribution between the fibers shown in Fig. 4(b) shows no significant difference between PVDF1 and PVDF2, PVDF2 and PVDF3, PVDF3 and PVDF4, PVDF 1 and PVDF3, PVDF 2 and PVDF4, and PVDF1 and PVDF4, with a significance value of $p \leq 0.001$. A p-value ≤ 0.05 indicates that there is no significant difference between the results. The degree of homogeneity in the diameter distribution of each PVDF sample was assessed by comparing the coefficient of variance (ε). A previous study considered that a nanofiber is homogeneous if its coefficient of variance is < 0.3 [37]. The investigation revealed that PVDF1 and PVDF2 exhibited a less homogeneous diameter distribution with values of 0.43 and 0.34, respectively, as compared with PVDF3 and PVDF4, which exhibited higher homogeneity with values of 0.22 and 0.10. The observed differences in homogeneity can be potentially attributed to current fluctuations during the solution electrospinning process [15]. The electrospinning machine possibly lacked a current fluctuation control system, leading to variations in the nanofiber diameters. Additionally, applying a high voltage without readjustment can result in an imbalance between the precursor and the spinner, contributing to homogeneity differences [38].

3.3. Mechanical properties of the PVDF nanofiber membranes

Fig. 5 and Table 2 present the mechanical test data for the PVDF1, PVDF2, PVDF3, and PVDF4 nanofiber membranes. The analysis results show that increasing the PVDF concentration from 15 to 30% proportionally increased the tensile strength of the PVDF nanofiber membranes from 3.61 to 6.00 MPa. This was attributed to the denser and more regular molecular structure and an increased quantity of hydrogen bonds between the polymer chains [36]. This result is consistent with the results of the study by Zaarour et al. , which reported that increasing the PVDF concentration from 17 to 30% increased the tensile strength of the produced nanofibers from 1.4 to 3.3 MPa [39]. The influence of an increasing concentration correlates with the increase in strain, where the strain of PVDF1, which was 4.2%, increased to 8.11% (PVDF4) because the resistance of the nanofibers to deformation improved with the presence of more hydrogen bonds between the PVDF polymer chains [40]. This finding is consistent with the findings of the study by Teng et al. [23], which reported that an increasing concentration of PVDF increases the strain of the produced nanofiber membrane [23]. However, an increase in the PVDF concentration decreased the Young's modulus values of the PVDF1, PVDF2, PVDF3, and PVDF 4 nanofiber membranes from 85.95 to 73.98 MPa. This decrease can be attributed to the increasing rigidity and regularity of the bonds between the PVDF molecules owing to the very high PVDF concentration, hindering their

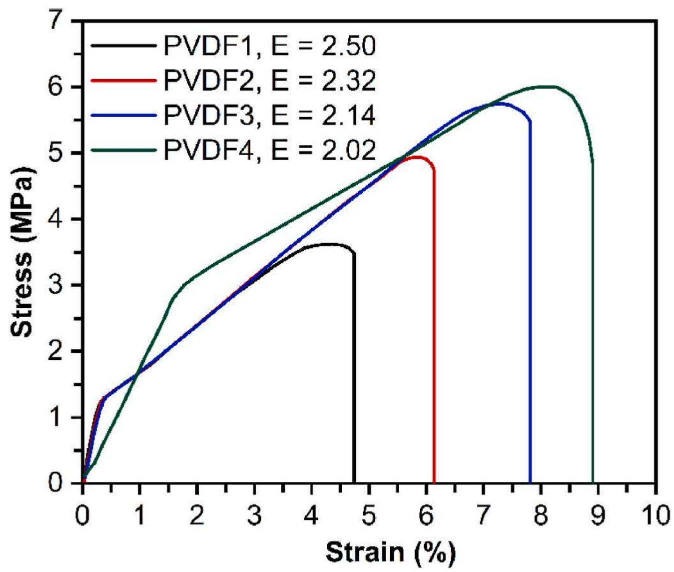

Fig. 5. Stress-strain curves of the (a) PVDF1, (b) PVDF2, (c) PVDF3, and (d) PVDF4 nanofiber membranes.

Table 2
The detail of mechanical properties of PVDF nanofiber membranes.

Sample	Tensile Strength (MPa)	Strain (%)	Young's modulus (MPa)
PVDF1	3.61	4.20	85.95
PVDF2	4.93	5.90	83.55
PVDF3	5.75	7.31	78.65
PVDF4	6.00	8.11	73.98

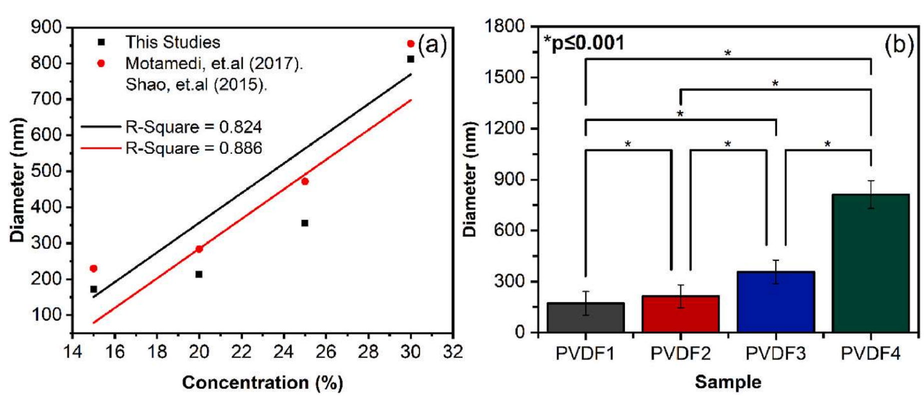

Fig. 4. (a) Comparison of the slope regression values of the PVDF nanofibers produced in this study and those produced in other studies, and (b) significance analysis of the diameter distribution between the (a) PVDF1, (b) PVDF2, (c) PVDF3, and (d) PVDF4 nanofibers.

smooth movement and flexibly. This result agrees with the morphological analysis described earlier. The tensile strength of PVDF varies depending on its concentration and application. For example, the tensile strength ranges for wound healing, water filter, and mask applications are 2.0–3.5 [40], 3.43–14.34 MPa [41], and 0.7–1.1 MPa [42], respectively.

3.4. FTIR analysis of the PVDF nanofiber membranes

PVDF is a multiphase polymer that crystallizes into different structural polymorphs. PVDF has superior characteristics, such as piezoelectricity and pyroelectricity, due to its inter-polarized structure of hydrogen (δ^+) and fluorine (δ^-) [43]. The PVDF structure can exist in at least three phases, namely, the α, β, and γ phases, and each phase is composed of the molecular conformation shown in Fig. 6(a). The chain conformation in the α phase is designated as a twisted trans-gauche (TGTG′) (T-trans, G-gauche$^+$, G-gauche$^-$) conformation, which is paraelectric because the dipole moment is neutralized as a result of the anti-parallel packing of the molecular chains in the unit cell of the α-phase crystal [39,44]. In the transplant zigzag chain conformation (TTTT) in β-phase PVDF, the polymer chains are arranged in one orientation, which allows all the dipoles to be layered and aligned along the crystallographic axes by spontaneous polarization of the unit cell [45]. Consequently, the PVDF β-phase exhibits ferroelectric properties. The conformation for the γ phase is TTTGTTTG', that is, T_3GT_3G', which is a conformation between the α and β phases. All individual chain dipoles in the unit cell are parallel, resulting in a nonzero dipole moment and polarity [46]. The PVDF phase differences can be determined using FTIR and XRD, which show significant peaks with similar intensities, experimental patterns, and spectra. FTIR spectroscopy was used to analyze the chemical structure and phase stabilization of the PVDF nanofiber samples. Fig. 6(b) shows the polarized FTIR spectra, in which enhanced absorption peaks appear when the molecular vibrations and electric field components of the incident FTIR are aligned. This study successfully observed the phase characteristics of the PVDF nanofiber membranes by analyzing the resulting wavenumber peaks. Table 3 lists the peak details of the resulting wavenumbers of the PVDF nanofibers.

The FTIR data was subsequently compared with the wavenumbers of the phases of PVDF; in particular, the data was compared with the three primary phases of PVDF (α, β, and γ poises) reported in previous studies [49,50]. The wavenumbers of the PVDF nanofiber membranes are compared in Fig. 6(b). Table 4 lists the wavenumbers corresponding to the three phases. PVDF1 exhibited a markedly strong phase signal with intensive peaks at 879 and 1072 cm $^{-1}$, which were attributed to the α phase, a characteristic peak at 1240 cm $^{-1}$, which was attributed to the γ phase, and peaks at 840, 1180, 1273, and 1401 cm $^{-1}$, which were

Table 3
Wavenumber of FTIR analysis of PVDF Nanofiber Membranes.

Sample	Wavenumber (cm^{-1})
PVDF1	1401, 1273, 1240, 1180, 1072, 879, 840
PVDF 2	1401, 1274, 1240, 1181, 1072, 877, 840
PVDF 3	1401, 1272, 1240, 1181, 1072, 878, 840
PVDF 4	1401, 1273, 1240, 1181, 1072, 878, 840

Table 4
Wavenumber of Phase α, β, γ. of PVDF.

Phase	Wavenumber (cm^{-1})	References
α	976, 878, 796, 614	[40,55–57]
β	1401, 1274, 1181, 840	[40,55–58]
γ	1240, 812, 776	[40,52,56]

Table 5
Peak number of Phase α, β, γ. of XRD PVDF.

Phase	Peaks number (°)	References
α	17.10, 17.31,17.32, 18.10, 18.6, 18.84, 20.10, 26.80	[56,57,64,65]
β	20.30, 20.40, 20.50, 20.60, 20.82, 22.28, 22.34, 36.44, 36.60	[56,57,64,65]
γ	18.70, 20.46, 20.52, 26.90, 38.16, 38.20	[56,57,64,65]

attributed to the β phase [51,52]. PVDF2 exhibited a characteristic γ-phase peak at 1181 cm $^{-1}$, an exclusively weakened α-phase peak at 1240 cm $^{-1}$, and an exclusively strengthened β-phase peak at 1274 cm $^{-1}$ [30,53]. The bands at 842, 1181, 1272, and 1401 cm $^{-1}$ in the PVDF3 spectrum are characteristic of the β phase, and the bands at 878, 1072, and 1240 cm $^{-1}$ are characteristic of the α and γ phases [53,54]. Moreover, the PVDF4 spectrum indicates a higher amount of the β phase with lower intensity peaks, although the typical α- and γ-phase peaks can still be observed [53] (see Table 5).

The difference between these phases is caused by dipolar interactions between C=O and CH_2-CF_2. The greater the concentration of PVDF, the greater the PVDF chain owing to the presence of strong hydrogen bonds, which in turn form an increasingly polar phase [59,60]. At higher concentrations, the dipole CH_2-CF_2 in PVDF becomes more densely packed due to dipole interactions and the hydrogen bonds at the intermolecular interface, facilitating the precipitation of the β phase [30,45]. In contrast, low concentrations of PVDF results in an insufficient force orientation to promote dipole packing, leading to the molecules being packed into the more thermodynamically stable α phase, mainly because

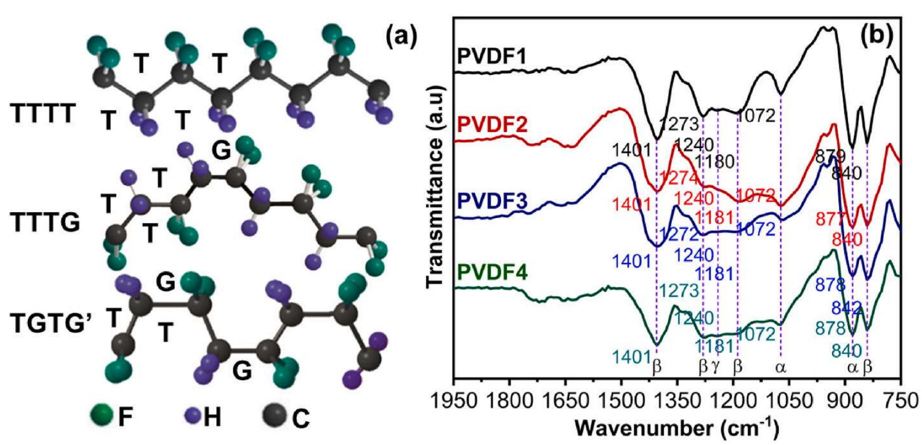

Fig. 6. **(a)** Schematic illustrations of the molecular chain conformations of the α, β, and γ phases of PVDF [47,48], and **(b)** FTIR spectra of the PVDF1, PVDF2, PVDF3, and PVDF4 nanofibers.

of the relatively high external energy during the drying process [30]. The three PVDF phases have piezoelectric properties, with the β phase contributing the most to piezoelectricity [30,39]. It should be noted that piezoelectricity can increase fiber elasticity [39,40]. Based on the FTIR results, PVDF1, PVDF2, PVDF3, and PVDF4 fibers can be applied in various ways based on their piezoelectric properties, including wound healing [28], water filters [61], and masks [62], all of which emphasize fiber elasticity.

3.5. XRD analysis of the PVDF nanofiber membranes

XRD has been extensively used to understand the structural modification of polymers by analyzing the position, shift, and expansion of diffraction peaks in the XRD patterns of each polymer sample [32,63]. Fig. 7(a) shows the XRD patterns of the PVDF nanofibers. PVDF nanofibers can crystallize in at least three polymorphs (α, β, and γ), with each phase having a different crystal structure. Tabel 5 lists the XRD peak data of the three polymorphs of PVDF nanofibers in previous studies.

The obtained data were used to comparatively analyze the XRD patterns of PVDF1, PVDF2, PVDF3, and PVDF4. These patterns were compared with the peak number data obtained from the XRD results. PVDF1 was mainly comprised of the α phase, as indicated by the presence of the three peaks at 17.30, 18.84, and 26.80°, which correspond to the (100), (020), and (021) planes of the monoclinic α phase crystals [45,56]. The two peaks at 20.52 and 38.20° correspond to the (110/101) and (110) planes of the γ phase of the monoclinic crystals [56,65]. Moreover, the three peaks at 20.82, 22.34, and 36.44° correspond to the reflections of the (110), (200), and (201) crystal planes of the orthorhombic β-phase [59,60]. The crystalline phase of the PVDF2 nanofibers exhibited peaks at 17.32, 18.84, 20.52, and 38.16°, corresponding to diffraction in the (100), (020), (110/101), (110) planes of the α and γ phases of the monoclinic crystals [21,39]. Moreover, the strong peak at 20.86° and the moderate peaks at 22.34 and 36.44° correspond to the (110), (200), and (101) planes of the orthorhombic β-phase crystals [44]. The PVDF3 nanofibers exhibited an intensive peak at 20.46°, confirming the excision of the γ phase in the (110/101) plane of the γ phase of the monoclinic crystals [66]. The β phase peak could still be observed at 22.28°, thus confirming the existence of an orthorhombic β phase in the (200) plane. The PVDF4 nanofibers did not exhibit α and γ phase peaks; however, a weak diffraction peak was still observed at 20.3°, which corresponds to the (110/200) plane of the orthorhombic β-phase crystals [67]; therefore, the β phase was dominant in the PVDF sample.

The findings from the XRD comparison agree with the FTIR data, which indicated a phase shift with an increasing PVDF concentration in the sample with a prevalence of the β phase. The β phase strengthened and the α and γ phases weakened with an increasing PVDF concentration. This outcome is due to the higher PVDF nanofiber concentrations resulting in the coexistence of α, β, and γ phases owing to the solute-solvent interaction during the electrospinning process. This interaction alters the activation energy imparted to the nanofiber samples when determining the solution parameters during electrospinning [40]. The phase data obtained from this analysis provides valuable insights into the crystallization level of the PVDF nanofibers. Fig. 7(a) shows the unit cells of the α, β, and γ phases of PVDF. The molecular structure of PVDF consists of carbon, hydrogen, and fluorine atoms. The packing of molecules into different shapes results in different crystal structures [40]. The α phase is non-polar, whereas the β and γ phases are polar. Although all three phases were observed in this study, the β phase grew at the expense of the other phases [68]. The β-phase crystals with all the transformations can form orthorhombic cells, producing the most polar phase among the crystal structures In contrast, the non-polar α phase can form monoclinic cells with a rectangular lattice. Notably, the bonding of the polar and non-polar phases determines the degree of piezoelectricity that the nanofibers can produce because of the presence of a dipole moment [39]. Fig. 7(b) illustrates that the non-polar phase is formed from intermolecular bonds in one element, which explains its dipole moment of zero [52,61]. The polar phase is formed from the molecular bonds between its unit elements; therefore, it has a dipole moment that is not equal to zero [52]. Therefore, it can be concluded that all the PVDF nanofibers produced in this study have good piezoelectric properties.

The XRD data enabled the crystallization level to be determined. The XRD data from all samples was deconvoluted to assess the crystallinity percentage (%χC) in the nanofibers [69]. A comparative XRD pattern served as a reference for demodulation, revealing five distinct weak peaks at 17.32, 18.84, 22.34, 26.80, and 38.16° and two strong peaks at 20.52 and 20.82°. The relatively intense crystal peaks were characterized by a decreased intensity and/or shifting of the diffraction peaks. This phenomenon can be associated with a decrease in the polymer crystallinity, and the %χC values of PVDF1, PVDF2, PVDF3, and PVDF4 were 58.45, 56.67, 54.67, and 52.67%, respectively. Zhu et al. reported that PVDF possesses no more than 50–60% crystallinity owing to the small size of the fluorine atom and its vinylidene structure. This results in a transition from a fully crystalline state to a semicrystalline state

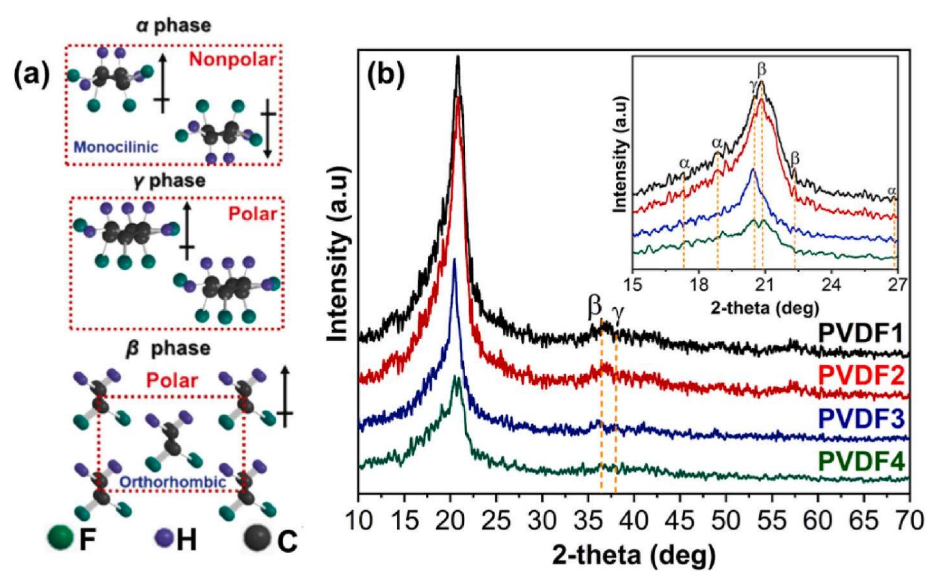

Fig. 7. (a) Schematic illustrations of the α, β, and γ phases of the PVDF unit cells [48] (b) XRD patterns of the PVDF1, PVDF2, PVDF3, and PVDF4 nanofibers. Inset shows the magnified pattern from 15 to 27°.

[70]. Crystal formation is hindered during the electrospinning process because of the rapid evaporation of the solvent, which causes the polymer chains to remain unaligned, and the intermolecular interactions lead to the formation of a less-organized polymer matrix [32].

4. Conclusion

Fine homogenous PVDF nanofiber membranes were successfully produced and characterized in this study. The results showed that increasing the PVDF concentration affected the physicochemical and mechanical properties of the nanofibers. The SEM analyses indicated that the fiber diameter increased with an increasing PVDF concentration, resulting in an average PVDF nanofiber diameter range of 172–812 nm. This increase in diameter positively influenced the tensile strength of the nanofiber membranes, which increased from 3.61 to 6.00 MPa. The FTIR results revealed the presence of various phases, namely the α, β, and γ phases, with the β phase being dominant. The XRD analyses demonstrated a semi-crystalline structure with crystallinity values of 58.45–52.67%. The findings of this study indicate that the PVDF nanofiber membranes are promising candidates for application in wound healing, water filtration, and mask production. This study provides valuable insights into the evolution of the characteristics of PVDF nanofiber membranes with an increasing PVDF concentration, making them suitable for use in various composite polymer applications.

CRediT authorship contribution statement

Ida Sriyanti: Funding acquisition, Project administration, Supervision, Writing – review & editing. **Rafli Fandu Ramadhani:** Conceptualization, Data curation, Methodology, Writing – original draft. **Muhammad Rama Almafie:** Formal analysis, Investigation. **Meutia Kamilatun Nuha Ap Idjan:** Formal analysis, Investigation. **Edi Syafri:** Formal analysis. **Indah Solihah:** Writing – original draft. **Muhammad Rudi Sanjaya:** Formal analysis, Investigation. **Jaidan Jauhari:** Formal analysis, Investigation. **Ahmad Fudholi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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